Orlicz spaces and the uncertainty principle

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The uncertainty principle asserts that a function and its Fourier transform cannot both be sharply localized: the product of their support sizes must be large. Recently, Iosevich and Mayeli refined this principle for functions whose Fourier transform is supported on specific subsets of $\mathbb{Z}_N^d$, using restriction theory in L^p spaces.

In this talk, we will extend their ideas to Orlicz spaces, which generalize L^p spaces and allow finer control over growth conditions of functions. We will introduce (Φ, Ψ) restriction estimates and use them to establish sharper uncertainty principles and new results related to Bourgain's Λ_p sets. Finally, we will discuss an application of these results to the exact recovery problem, showing how the Orlicz space framework recovers and extends the classical results of Iosevich and Mayeli in the Lebesgue setting. This presentation is based on joint work with Alex Iosevich, Isaac Li, and Zhihe Li.